

Exercise 6.12

Consider a hydrogen atom whose state at time $t = 0$ is given by

$$\Psi(\vec{r}, 0) = \frac{1}{\sqrt{2}}\phi_{300}(\vec{r}) + \frac{1}{\sqrt{3}}\phi_{311}(\vec{r}) + \frac{1}{\sqrt{6}}\phi_{322}(\vec{r}).$$

- (a) What is the time-dependent wave function?
- (b) If a measurement of the energy were carried out, what values could be found and with what probabilities?
- (c) Repeat part (b) for $\hat{L}^2$ and $\hat{L}_z$. That is, if a measurement of $\hat{L}^2$ and $\hat{L}_z$ were carried out, what values could be found and with what probabilities?

(a) Lets recall some notation:
Here the notation is ϕ_{nlm} but in your notes it is ψ_{nlm}

$$\vec{\psi}(\vec{r}, 0) = \frac{1}{\sqrt{2}}\psi_{300}(\vec{r}) + \frac{1}{\sqrt{3}}\psi_{311}(\vec{r}) + \frac{1}{\sqrt{6}}\psi_{322}(\vec{r})$$

note:

$$\psi_{nlm} \rightarrow n=3$$

for all 3, thus ψ (and all ψ_{3lm}) are Energy Eigen functions of the Hamiltonian

$$\hat{H}(\psi_{3lm}) = E_3\psi_{3lm}$$

- Why is this relevant? Well, we know to find a state $\vec{\psi}(\vec{r}, 0)$ at a later time we evolve with the operator

$$e^{-i\hat{H}t} \Rightarrow \psi(\vec{r}, t) = e^{-i\hat{H}t} \psi(\vec{r}, 0)$$

Recap: Given that you've been tortured by boring thermo lets recap where this operator comes from:

Schrödinger Eq:

$$i\partial_t \psi(x) = (\hat{H}\psi)(x)$$

This has a "formal" solution of the form: $\frac{d\psi}{dt} = -i\hat{H}\psi$

$$\text{or } \int_{\psi_0}^{\psi} \frac{d\psi}{\psi} = \int_0^t -i\hat{H} dt$$

$$\ln(\psi) \Big|_{\psi_0}^{\psi} = -i\hat{H}t$$

$$\ln(\psi) - \ln(\psi_0) = -i\hat{H}t$$

$$\Rightarrow \ln(\psi/\psi_0) = -i\hat{H}t$$

$$\psi/\psi_0 = e^{-i\hat{H}t}$$

Here

$$\psi_0 \equiv \psi(\vec{r}, 0)$$

Assumes $\hat{H}$ does not depend on t

or

$$\psi(\vec{r}, t) = e^{-i\hat{H}t} \psi(\vec{r}, 0)$$

→ Strictly speaking this "proof" isn't quite right but it does give you a "reliable" way of figuring out where it comes from.

• We know that $\psi(\vec{r}, 0)$ is an eigenvector of $\hat{H}$, so

$$e^{-i\hat{H}t} \psi(\vec{r}, 0) = e^{-iE_3t} \psi(\vec{r}, 0)$$

$$E_n \equiv -13.6 \text{ eV } \frac{1}{n^2}$$

Time dependent wave fn:

$$\psi(\vec{r}, t) = e^{-iE_3t} \left(\frac{1}{\sqrt{2}} \psi_{300}(\vec{r}) + \frac{1}{\sqrt{3}} \psi_{311}(\vec{r}) + \frac{1}{6} \psi_{322}(\vec{r}) \right)$$

→ Note: we are using natural units so $\hbar = 1$. Otherwise the time ev. operator is $-i\hat{H}t/\hbar$

(b) Since ψ and all three u 's are eigen functions of $\hat{H}$ with (eigen value) Energy E_3 : the system can only have energy E_3 so the probability is 1.

(c) Recall:

$$\begin{array}{lcl}
 & \xrightarrow{\text{Energy}} & \hat{H}(u_{nlm}) = E_n(u_{nlm}) \\
 u_{nlm} & \xrightarrow{\text{z-component of Ang. M.}} & \hat{L}_z(u_{nlm}) = \hbar m(u_{nlm}) \\
 & \xrightarrow{\text{Total Ang. momenta}} & \hat{L}^2(u_{nlm}) = \hbar^2 l(l+1)(u_{nlm})
 \end{array}$$

so we see we have different possible

Note, from

$$\psi(\vec{r}, 0) = \frac{1}{\sqrt{2}} u_{300} + \frac{1}{\sqrt{3}} u_{311} + \frac{1}{\sqrt{6}} u_{322}$$

$\left. \begin{array}{l} l=0 \\ l=1 \\ l=2 \end{array} \right\} \text{three possibilities}$

Recall: ψ is normalized since
 $\langle \psi | \psi \rangle = \frac{1}{2} + \frac{1}{3} + \frac{1}{6} = \frac{3}{6} + \frac{2}{6} + \frac{1}{6} = 1 \checkmark$
 thus, the coefficients are the square roots of the probabilities

$l=0$: u_{300}

$$P = |\langle u_{300} | \psi \rangle|^2 = \left(\frac{1}{\sqrt{2}} \right)^2 = \frac{1}{2}$$

→ The value $\hbar^2 0 \cdot (0+1) = 0$ for a measurement of $\vec{L}^2$ will happen with a probability of $\frac{1}{2}$

$l=1$: ψ_{311}

$$P_1 = |\langle 311 | \psi \rangle|^2 = \left(\frac{1}{\sqrt{3}}\right)^2 = \underline{\frac{1}{3}}$$

The value $\hbar^2 1(1+1) = 2\hbar^2$ for a measurement of $\vec{L}^2$ will occur w prob. $\frac{1}{3}$

$l=2$: ψ_{322} :

$$P_2 = |\langle 322 | \psi \rangle|^2 = \left(\frac{2}{\sqrt{6}}\right)^2 = \underline{\frac{2}{3}}$$

• Do the exact same for m

$$m=0 : \psi_{300}$$

$$m=1 : \psi_{311}$$

$$m=2 : \psi_{322}$$

An extra note:

The problem above is a total eigenfunction of $\hat{H}$ since

$$\hat{H} \psi(\vec{r}, 0) = E_3 \psi(\vec{r}, 0)$$

This is not always the case. Suppose

$$\psi'(\vec{r}, 0) = \sqrt{\frac{2}{15}} \psi_{200} + \frac{1}{\sqrt{15}} \psi_{311} + \frac{2}{\sqrt{3}} \psi_{422}$$

$\downarrow$
 E_2

$\downarrow$
 E_3

$\downarrow$
 E_4

• Each of these are different eigen functions of $\hat{H}$ so the time dependent state will be

$$\begin{aligned}\psi(\vec{r}, t) &= e^{-i\hat{H}t} \psi(\vec{r}, 0) \\ &= \sqrt{\frac{7}{15}} e^{-iE_2 t} \psi_{200} + \frac{1}{\sqrt{5}} e^{-iE_3 t} \psi_{311} + \frac{1}{\sqrt{3}} e^{-iE_4 t} \psi_{422}\end{aligned}$$

prob of Measuring energies E_i :

$$\begin{aligned}i) P(E_2) &= \frac{7}{15} \\ ii) P(E_3) &= \frac{1}{5} \\ iii) P(E_4) &= \frac{1}{3}\end{aligned} \quad \left. \vphantom{\begin{aligned} i) P(E_2) &= \frac{7}{15} \\ ii) P(E_3) &= \frac{1}{5} \\ iii) P(E_4) &= \frac{1}{3} \end{aligned}} \right\} \frac{7}{15} + \frac{1}{5} + \frac{1}{3} = 1 \quad \checkmark$$

